Advanced Python Technical Documentation for RAG and AI Systems

This document is a comprehensive technical reference focused on Python engineering concepts, backend architecture, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), LLM pipelines, APIs, vector databases, asynchronous processing, deployment workflows, containerization, embeddings, transformers, and production-grade AI systems. The content is intentionally detailed and verbose so that the document can be used for: - Retrieval-Augmented Generation (RAG) - Embedding generation - Semantic chunking - Vector search experimentation - LangChain and LangGraph pipelines - Production AI testing - Long-context retrieval evaluation

Chapter 1: Python Runtime Internals

Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead.

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Example Code:

```
SELECT users.name, orders.total
FROM users
JOIN orders ON users.id = orders.user_id
WHERE orders.total > 1000;
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 2: Object-Oriented Programming

Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization.

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 3: Concurrency and AsyncIO

Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets. Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets. Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets.

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Example Code:

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from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
tokens = tokenizer("Hello world")
print(tokens)
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
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Chapter 4: REST API Development

REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth. REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth. REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth.

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Example Code:

```
docker compose up -d
docker compose logs
docker compose down
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
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Vector Database	Semantic retrieval	High
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Chapter 5: Database Integration

SQLAlchemy provides ORM functionality for relational databases. Connection pooling, migrations, indexes, joins, and transaction management are critical concepts in backend engineering. SQLAlchemy provides ORM functionality for relational databases. Connection pooling, migrations, indexes, joins, and transaction management are critical concepts in backend engineering. SQLAlchemy provides ORM functionality for relational databases. Connection pooling, migrations, indexes, joins, and transaction management are critical concepts in backend engineering.

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Example Code:

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from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

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Chapter 6: Docker and Containerization

Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments. Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments. Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments.

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Tokenizer	Convert text into tokens	Medium
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Chapter 7: Kubernetes

Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems. Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems. Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems.

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Chapter 8: Natural Language Processing

NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models. NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models. NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models.

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Chapter 9: Embeddings and Vector Databases

Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search. Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search. Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search.

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Component	Purpose	Complexity
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Chapter 10: Retrieval-Augmented Generation

RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding. RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding. RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding.

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```

Component	Purpose	Complexity

Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
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Chapter 11: Transformers and Attention

Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens. Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens. Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens.

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Component	**Purpose**	**Complexity**
Tokenizer	Convert text into tokens	Medium
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Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 12: Inference Optimization

Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration.

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Example Code:

```
docker compose up -d
docker compose logs
docker compose down
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Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
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Chapter 13: LLMOps

LLMOps focuses on monitoring, observability, prompt management, evaluation, deployment, and scalability of AI systems. LLMOps focuses on monitoring, observability, prompt management, evaluation, deployment, and scalability of AI systems. LLMOps focuses on monitoring, observability, prompt management, evaluation, deployment, and scalability of AI systems.

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Example Code:

```
SELECT users.name, orders.total
FROM users
JOIN orders ON users.id = orders.user_id
WHERE orders.total > 1000;
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 14: Python Runtime Internals

Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead.

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Example Code:

```
import asyncio

async def fetch_data():
    await asyncio.sleep(1)
    return "completed"

asyncio.run(fetch_data())
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 15: Object-Oriented Programming

Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization.

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 16: Concurrency and AsyncIO

Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets.

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Example Code:

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```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 17: REST API Development

REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth. REST APIs

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 18: Database Integration

SQLAlchemy provides ORM functionality for relational databases. Connection pooling, migrations, indexes, joins, and transaction management are critical concepts in backend engineering. SQLAlchemy provides ORM functionality for

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Example Code:

```
SELECT users.name, orders.total
FROM users
JOIN orders ON users.id = orders.user_id
WHERE orders.total > 1000;
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 19: Docker and Containerization

Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments. Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments. Docker packages applications and dependencies into containers. Docker Compose orchestrates multi-container systems. Containers provide consistency across development and production environments.

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Example Code:

```
docker compose up -d
docker compose logs
docker compose down
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 20: Kubernetes

Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems. Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems. Kubernetes manages container orchestration using Pods, Deployments, Services, ConfigMaps, Secrets, and Ingress controllers. Horizontal Pod Autoscaling supports scalable systems.

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 21: Natural Language Processing

NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models. NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models. NLP converts human language into numerical representations. Tokenization, embeddings, attention mechanisms, and transformer architectures form the foundation of modern language models.

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Example Code:

```
import asyncio

async def fetch_data():
    await asyncio.sleep(1)
    return "completed"

asyncio.run(fetch_data())
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 22: Embeddings and Vector Databases

Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search. Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search. Embeddings are dense numerical vectors representing semantic meaning. Vector databases such as FAISS, Pinecone, and ChromaDB enable semantic similarity search.

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Example Code:

```
SELECT users.name, orders.total
FROM users
JOIN orders ON users.id = orders.user_id
WHERE orders.total > 1000;
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 23: Retrieval-Augmented Generation

RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding. RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding. RAG systems combine retrieval systems with language models. Documents are chunked, embedded, stored in vector databases, and retrieved during inference to improve grounding.

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embedded, stored in vector databases, and retrieved during inference to improve grounding.

Example Code:

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
tokens = tokenizer("Hello world")
print(tokens)
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 24: Transformers and Attention

Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens. Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens. Transformers rely on self-attention mechanisms. Multi-head attention enables models to learn complex contextual relationships between tokens.

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 25: Inference Optimization

Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration.

Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration. Inference optimization includes quantization, batching, caching, tensor parallelism, and GPU acceleration.

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

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    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 26: LLMOps

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Example Code:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 27: Python Runtime Internals

Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead. Python executes code using the CPython interpreter. Source code is converted into bytecode and executed by the Python Virtual Machine. Memory management is handled through reference counting and garbage collection. Python objects contain metadata such as type information and reference counts. Dynamic typing enables flexibility but introduces runtime overhead.

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Example Code:

```
import asyncio

async def fetch_data():
    await asyncio.sleep(1)
    return "completed"

asyncio.run(fetch_data())
```

Component	Purpose	Complexity

Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 28: Object-Oriented Programming

Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization. Python supports classes, inheritance, polymorphism, abstraction, and encapsulation. Every entity in Python is an object. The method resolution order determines inheritance traversal. Dunder methods enable operator overloading and protocol customization.

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Example Code:

```
docker compose up -d
docker compose logs
docker compose down
```

Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
Embedding Model	Generate semantic vectors	High
Vector Database	Semantic retrieval	High
LLM	Generate responses	Very High

Chapter 29: Concurrency and AsyncIO

Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets. Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets. Concurrency in Python can be implemented using threading, multiprocessing, and asynchronous IO. AsyncIO enables cooperative multitasking using event loops, coroutines, await expressions, and asynchronous sockets.

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Component	Purpose	Complexity
Tokenizer	Convert text into tokens	Medium
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Vector Database	Semantic retrieval	High
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Chapter 30: REST API Development

REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth. REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth. REST APIs are commonly built using Flask and FastAPI. APIs expose resources using HTTP methods such as GET, POST, PUT, PATCH, and DELETE. Authentication can be implemented using JWT tokens and OAuth.

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